

Isaac Mubiru

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PROFESSIONAL SUMMARY

Full-stack software engineer with 4+ years building user-facing products and backend services. Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, cutting \$5K/month in infrastructure costs. Proficient in React, TypeScript, Node.js, and cloud platforms (Firebase, GCP, AWS). Experienced designing reusable abstractions, component libraries, and real-time data pipelines. Comfortable navigating ambiguous requirements and cross-functional collaboration with direct CTO reporting.

CORE COMPETENCIES

React & TypeScript

Full-Stack Development

APIs & Backend Services

Dashboard & UI Systems

Performance Optimization

Cloud Migrations (GCP/Firebase)

Real-Time Data Pipelines

Microservices Architecture

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led **zero-downtime migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP, saving \$5,000/month in infrastructure costs
- Built **real-time two-way sync pipeline** between MongoDB and Firestore using Node.js and Google Pub/Sub, supporting parallel production traffic during migration
- Developed **reusable component libraries** and shared abstractions used across multiple applications for consistent UI and backend patterns
- Built a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping feature generating \$40 additional revenue per order

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Built migration scripts to read data from MongoDB and write to Firestore after complex processing logic
- Wrote documentation for engineering team on new Firebase SDKs (Node.js and Python)
- Trained operations team on Firebase Dashboard and backup procedures

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backend services for DeFi applications using Web3 and Node.js, reporting directly to CTO
- Delivered **React UIs from Figma designs** under tight timelines for demanding clients

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led **performance optimization** of a PWA dashboard delivering real-time stock market prices from Firebase backend
- Built ad-hoc jobs to query 2M+ Firestore records, keeping customer prices current from Morningstar APIs
- Added responsive UI pages using React, HTML, and Tailwind CSS
- Extended backend APIs using Firebase Cloud Functions and Express.js

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected **microservices backend** using Apache Kafka, Docker, and Kubernetes for case management application
- Developed cross-platform mobile app using React Native and Expo (iOS & Android) with Firebase push notifications
- Built USSD service communicating with Node.js backend via gRPC for legal aid providers

PROJECTS

Real-time Pub/Sub Sync Pipeline

Built live MongoDB to Firestore synchronization pipeline supporting parallel production traffic during zero-downtime migration at MTailor

Node.js, Google Pub/Sub, MongoDB, Firestore, GCP

Healthcare WhatsApp Chatbot

Integrated OpenAI and Google Vision for prescription OCR via WhatsApp Business API, automating healthcare intake workflows

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Marketplace Recommendation Engine

Built collaborative and content-based filtering engine in MongoDB for service provider matching

Node.js, MongoDB, Machine Learning

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#)

2018 – 2022

Kampala, Uganda

CERTIFICATIONS

No certifications listed

SKILLS

Frontend: React, TypeScript, JavaScript, HTML, CSS, Tailwind, React Native **Backend:** Node.js, Express.js, Firebase Cloud Functions, Python, Go

Databases: MongoDB, Firestore, SQL **Cloud:** GCP, Firebase, AWS (S3, EC2, EBS)

Infrastructure: Docker, Kubernetes, Apache Kafka, Google Pub/Sub, gRPC **Tools:** Git, Linux, ffmpeg